

JEE MAIN REVISION MASTER: MATHEMATICAL REASONING OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 121 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Mathematical Reasoning — Core Principles & Governing Laws	104 PYQs	Core Principle
High	Mathematical Reasoning — Standard Calculations & Problem Archetypes	11 PYQs	Quantitative Analysis
High	Mathematical Reasoning — Applied Cases, Reactions & Traps	8 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mathematical Reasoning — Core Principles & Governing Laws	$\therefore \sim[p \vee (\sim(p \wedge q))] = \sim p \wedge (p \wedge q), \quad (p \wedge \sim q) \Rightarrow (p \Rightarrow \sim q) = E, \quad p \Rightarrow (p \wedge q) = I$
Mathematical Reasoning — Standard Calculations & Problem Archetypes	$q = T$ and $(\theta = 240^\circ)$, is True, is equal to _____.
Mathematical Reasoning — Applied Cases, Reactions & Traps	$S_2: (q \Rightarrow p) \Rightarrow ((\sim p) \wedge q), \quad S_1: ((\sim p) \vee q) \vee ((\sim p) \vee r)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mathematical Reasoning — Core Principles & Governing Laws (Very High Yield — 104 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Mathematical Reasoning.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Mathematical Reasoning PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Negation of $p \wedge (q \wedge \sim(p \wedge q))$ is :..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "The converse of $((\sim p) \wedge q) \Rightarrow r$ is..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Negation of $(p \Rightarrow q) \Rightarrow (q \Rightarrow p)$ is :..." [Ans: N/A]

3.2 Mathematical Reasoning — Standard Calculations & Problem Archetypes (High Yield — 11 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Negation of $p \wedge (q \wedge \sim(p \wedge q))$ is :..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (Numerical): "The number of ordered triplets of the truth values of p, q and r such that the truth value of the statement $(p \vee q) \wedge (p \vee r) \wedge (q \vee r)$ is a c..." [Ans: 7]
- [2023 Evening Shift] Q#2 (MCQ): "Among the statements (S1) : $(p \Rightarrow q) \vee ((\sim p) \wedge q)$ is a tautology (S2) : $(q \Rightarrow p) \Rightarrow ((\sim p) \wedge q)$ is a c..." [Ans: N/A]

3.3 Mathematical Reasoning — Applied Cases, Reactions & Traps (High Yield — 8 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#2 (MCQ): "Among the statements (S1) : $(p \Rightarrow q) \vee ((\sim p) \wedge q)$ is a tautology (S2) : $(q \Rightarrow p) \Rightarrow ((\sim p) \wedge q)$ is a c..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "The compound statement $(\sim(P \wedge Q)) \vee ((\sim P) \wedge Q) \Rightarrow ((\sim P) \wedge (\sim Q))$ is equivalent to..." [Ans: N/A]
- [2023 Morning Shift] Q#3 (MCQ): " $(S1)(p \Rightarrow q) \vee (p \wedge (\sim q))$ is a tautology (S2) $((\sim p) \Rightarrow (\sim q)) \wedge ((\sim p) \vee q)$ is a contradiction. Then..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

Negation of $p \wedge (q \wedge \sim(p \wedge q))$ is :

- (A) $(\sim(p \wedge q)) \wedge q$
- (B) $(\sim(p \wedge q)) \vee p$
- (C) $p \vee q$
- (D) $\sim(p \vee q)$

Correct Answer: None

Solution Analysis: The given statement is: $p \wedge (q \wedge \sim(p \wedge q))$ We want to find the negation of this statement. To do that, we negate the whole expression: $\sim(p \wedge (q \wedge \sim(p \wedge q)))$ Now, we will use De Morgan's laws to simplify the expression. De Morgan's laws state: $\sim(A \wedge B) \equiv \sim A \vee \sim B$ $\sim(A \vee B) \equiv \sim A \wedge \sim B$ Applying the first De Morgan's law to the outer conjunction of the negated expression: $\sim p \vee \sim(q \wedge \sim(p \wedge q))$ Now we need to address the inner conjunction $(q \wedge \sim(p \wedge q))$. To simplify this part of the expression, we again apply the first De Morg

Benchmark Problem 2: JEE Main 2023 (Online) 11th April Morning Shift (Q#1)

The number of ordered triplets of the truth values of p, q and r such that the truth value of the statement $(p \vee q) \wedge (p \vee r) \Rightarrow (q \vee r)$ is True, is equal to _____.

Correct Answer: 7

Solution Analysis: $p \ q \ r \ p \vee q \ p \vee r \ (p \vee q) \wedge (p \vee r) \ q \vee r \ (p \vee q) \wedge (p \vee r) \Rightarrow (q \vee r)$ T T T T T T T T T T F T T T T T F T T T T T F F T T T T T T F T F T F F T T F F T F T F T F F F F F F F T Hence, the total number of ordered triplets are 7.

Benchmark Problem 3: JEE Main 2023 (Online) 24th January Morning Shift (Q#2)

The compound statement $(\sim(P \wedge Q)) \vee ((\sim P) \wedge Q) \Rightarrow ((\sim P) \wedge \sim Q)$ is equivalent to

- (A) $((\sim P) \vee Q) \wedge \sim Q$
- (B) $(\sim Q) \vee P$
- (C) $((\sim P) \vee Q) \wedge ((\sim Q) \vee P)$
- (D) $(\sim P) \vee Q$

Correct Answer: None

Solution Analysis: $(\sim(P \wedge Q)) \vee ((\sim P) \wedge Q) \Rightarrow ((\sim P) \wedge \sim Q)$ $(\sim P \vee \sim Q) \vee (\sim P \wedge Q) \Rightarrow (\sim P \wedge \sim Q)$ $\Rightarrow (\sim P \vee \sim Q \vee \sim P) \wedge (\sim P \vee \sim Q \vee Q) \Rightarrow (\sim P \wedge \sim Q) \Rightarrow (\sim P \vee \sim Q) \wedge (T) \Rightarrow (\sim P \wedge \sim Q) \Rightarrow (\sim P \vee \sim Q) \Rightarrow (\sim P \wedge \sim Q) \Rightarrow \sim(\sim P \vee \sim Q) \vee (\sim P \wedge \sim Q) \Rightarrow (P \wedge Q) \vee (\sim P \wedge \sim Q) \Rightarrow (\sim P \vee Q) \wedge (P \vee \sim P)$